

# PROJECT DOCUMENTATION

## CONTRACT ANALYSIS & RISK ASSESSMENT BOT FOR SME'S

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## 1. INTRODUCTION

Legal contracts are a daily reality for Small and Medium Enterprises (SMEs). Whether it is hiring employees, leasing property, or signing vendor agreements, contracts define the rights, duties, and risks of a business. However, most contracts are written in complex legal language that is difficult for non-lawyers to understand. As a result, many SME owners sign agreements without fully knowing what they are agreeing to.

This project introduces a GenAI-powered Contract Analysis & Risk Assessment Bot designed specifically for SMEs. The system reads contracts, identifies risky or unfair clauses, and explains them in simple, business-friendly language. The goal is to help business owners understand contracts clearly and make informed decisions before signing.

## 2. PROBLEM STATEMENT

SMEs often face serious challenges when dealing with legal contracts. Most contracts are long, complex, and full of legal terminology that is hard to interpret. Important risks such as unlimited liability, hidden penalties, or one-sided termination clauses are often buried deep inside the document.

Because of this complexity, SMEs depend heavily on legal consultants, which can be expensive and time-consuming. Many small businesses cannot afford frequent legal reviews and end up taking avoidable risks. There is a strong need for an affordable, automated solution that explains contracts clearly and highlights risks without requiring legal expertise.

## 3. PROPOSED SOLUTION

The proposed solution is a **web-based AI system** that allows users to upload legal contracts and receive an easy-to-understand analysis. Once a contract is uploaded, the system breaks it into individual clauses and evaluates each one for potential risks.

The tool provides clause-by-clause explanations, classifies risk levels as Low, Medium, or High, and highlights clauses that may be unfavorable to SMEs. It also suggests safer, SME-friendly alternative wording wherever possible. Finally, the system generates a

downloadable PDF report that can be used for internal review or legal consultation.

#### 4. OBJECTIVES

The main objective of this project is to simplify legal contract understanding for small and medium business owners. The system aims to automatically detect legal and financial risks and present them in a way that is easy to understand.

Another important objective is transparency. The system does not just give a risk score but clearly explains *why* a clause is risky and *how* it can be improved.

The project also aims to support multilingual contracts, especially English and Hindi, while fully complying with the hackathon constraints.

#### 5. SCOPE OF THE PROJECT

##### **In Scope**

This system is designed to work with common contracts used by SMEs, including employment agreements, vendor contracts, lease agreements, partnership deeds, and service contracts. These contract types cover most of the legal documents small businesses deal with regularly.

##### **Out of Scope**

The system does not provide certified legal advice and does not interpret court judgments or case law. It also does not connect to external legal databases or government APIs. The goal is assistance and awareness, not replacing professional lawyers.

#### 6. SYSTEM ARCHITECTURE OVERVIEW

The system follows a modular NLP pipeline architecture, where each stage performs a specific task. The process starts with document upload and OCR for scanned files. The system then detects the language and normalizes the text.

After that, the contract type is identified, clauses are extracted, and important entities such as parties, dates, and amounts are detected. Each clause is analyzed for risk and ambiguity. Finally, the LLM generates simple explanations and suggestions, and a structured PDF report is created. This modular design makes the system easy to maintain, test, and extend.

## 7. KEY FUNCTIONAL FEATURES

The system automatically classifies the type of contract and extracts clauses and sub-clauses accurately. It identifies key entities such as parties involved, financial amounts, dates, and jurisdiction details. Each clause is analyzed to determine whether it represents an obligation, a right, or a restriction. Risk scoring is done at the clause level, allowing users to clearly see which sections are safe and which require attention. The system also detects missing clauses, ambiguous language, and high-risk elements like penalties, indemnities, non-compete clauses, and IP transfer terms.

## 8. TECHNOLOGY STACK

The frontend is built using Streamlit, providing a clean and interactive user interface. Tesseract OCR is used to process scanned documents. Natural Language Processing is handled using spaCy and NLTK, while Scikit-learn is used for machine learning-based classification.

Large Language Models such as GPT-4 or Claude 3 are used only for explanation and suggestion generation. Testing is done using PyTest, and audit logs are stored locally in JSON format. PDF reports are generated without using any external services or APIs.

## 9. OCR & MULTILINGUAL SUPPORT

Many SMEs deal with scanned or photographed contracts. To handle this, the system uses OCR to convert scanned documents into text. The system automatically detects whether the contract is written in English or Hindi.

For Hindi contracts, the content is internally normalized to English for analysis. However, all explanations provided to the user are written in simple business English to ensure clarity and consistency.

## 10. RISK ASSESSMENT METHODOLOGY

Each clause is evaluated using a combination of rule-based logic and NLP techniques. The system looks for risky keywords, one-sided obligations, vague language, and imbalance between parties' rights and duties.

Based on this analysis, each clause is classified as Low, Medium, or High risk. A composite contract risk score is then calculated by combining all clause-level risks. This gives users a quick overall understanding of how risky the contract is.

## 11. TESTING & VALIDATION

To ensure reliability, the project includes a comprehensive testing framework. Automated tests verify clause extraction accuracy, risk scoring logic, OCR output quality, and entity recognition performance.

Decision engine tests ensure that risk classifications and explanations remain consistent. This testing approach helps maintain accuracy and trustworthiness of the system.

## 12. SECURITY & CONFIDENTIALITY

Security and privacy are core priorities of this project. All contracts are processed locally, and no data is sent to external servers. Sensitive files are excluded from version control using .gitignore.

Audit logs are maintained to ensure transparency and traceability without exposing contract content. This approach ensures confidentiality for SME users.

## 13. USE CASE SCENARIO

Consider an SME owner uploading a lease agreement. The system analyzes the document and identifies lock-in periods, penalty clauses, and unilateral termination rights.

High-risk clauses are clearly flagged, explained in simple language, and accompanied by safer alternative suggestions. A PDF risk report is generated, enabling the business owner to confidently negotiate better terms.

## 14. EXPECTED IMPACT

This system helps SMEs reduce legal and financial risk by making contracts transparent and understandable. It saves time and reduces dependency on expensive legal consultations.

By empowering business owners with clear insights, the tool supports smarter decision-making and more balanced agreements.

## 15. LIMITATIONS

The system is not a replacement for professional legal advice. Compliance checks are rule-based and do not reflect real-time legal changes. OCR accuracy may vary depending on document quality.

## 16. FUTURE ENHANCEMENTS

Future improvements include support for more regional languages, interactive clause comparisons, contract version comparison, and role-based dashboards. Secure cloud deployment with encryption can also be explored

## 17. CONCLUSION

The Contract Analysis & Risk Assessment Bot demonstrates how GenAI, OCR, and NLP can be combined to solve real-world legal challenges faced by SMEs. By translating complex legal language into simple, actionable insights, the system empowers businesses to make safer and more informed contractual decisions.